

WEEK 11

COLOR MEMORY NOTES

Cloud Computing + Sensor-Cloud + Fog/Edge/MEC/Cloudlet

Fast revision for tomorrow's exam

COLOR CODE

BLUE

Definition / core concept

GREEN

Features / benefits

YELLOW

Must-memorize facts

ORANGE

Comparison / selection clue

PURPLE

Memory trick

RED

Very high exam priority

Emergency Priority

- 1) NIST characteristics
- 2) SaaS/PaaS/IaaS
- 3) Deployment models
- 4) Sensor-Cloud
- 5) WSN vs Sensor-Cloud
- 6) MCC/Fog/Edge
- 7) Fog vs MEC vs Cloudlet
- 8) Virtualization

1. MASTER EXAM MAP

Identify the answer from the clue

Scale up/down quickly

->

Rapid Elasticity

Usage recorded + monitored

->

Measured Service

Use online software

->

SaaS

Build/deploy an app

->

PaaS

Need VM/OS/network/storage

->

IaaS

Cloud for one organization

->

Private Cloud

Private + Public together

->

Hybrid Cloud

One sensor serves many apps

->

Sensor-Cloud

Router/gateway near device

->

Fog Computing

Base-station/mobile edge

->

MEC

Data center in a box

->

Cloudlet

MASTER CHAIN

SORA -> BRMOR -> SPI -> PPHC -> EOS -> BCS -> Fog/Edge/MEC/Cloudlet

2. Cloud Computing - Core Idea

Start every 8-mark answer with a clean definition

EASY DEFINITION

Cloud computing provides convenient, on-demand network access to a shared pool of configurable computing resources such as servers, storage, networks and applications.

CORE IDEAS

- Shared pool of computing resources
- On-demand services
- High-level abstraction of computation and storage
- Rapid allocation and release
- Low management effort
- Can be accessed from different places and devices
- Supports service models and deployment models

CLOUD MEMORY

SORA = Shared - On-demand - Rapid - Anywhere

NIST VISUAL MODEL

5 Essential Characteristics + 3 Service Models + Deployment Models

3. Five NIST Essential Characteristics

Very high priority - learn the names and one line each

Broad Network Access

Resources available over network; supports heterogeneous clients.

Rapid Elasticity

Resources scale out/in rapidly and automatically.

Measured Service

Usage is recorded, monitored, controlled and optimized.

On-Demand Self-Service

Users get server time/storage automatically when needed.

Resource Pooling

Provider pools resources for multiple users via multitenancy.

NIST MEMORY

BRMOR = Broad - Rapid - Measured - On-demand - Resource Pooling | Think: Big Resources Move On Request

4. Service Models - SaaS / PaaS / IaaS

One of the most common 8-mark questions



SERVICE MEMORY

SPI = Software - Platform - Infrastructure | SaaS = USE | PaaS = BUILD | IaaS = MANAGE

5. Deployment Models

Public, Private, Hybrid, Community

Public

Open for general use. Low setup cost, scalable, pay-per-use.

Private

Used by one organization. More control, fewer security concerns, more maintenance.

Hybrid

Combination of two or more clouds; supports portability and load balancing.

Community

Shared by organizations with common security/compliance/jurisdiction concerns.

DEPLOYMENT MEMORY

PPHC = Public - Private - Hybrid - Community | Public = Everyone | Private = One org | Hybrid = Mix | Community = Common concern

6. Cloud Advantages + Components

Good 5-mark material and easy supporting points

BUSINESS ADVANTAGES

- Near-zero upfront infrastructure cost
- Real-time infrastructure availability
- Efficient resource utilization
- Usage-based costing
- Reduced time to market

GENERAL FEATURES

- Agility
- Ubiquitous access
- Multitenancy
- Dynamic load balancing
- Reliable and scalable
- Low cost and maintenance
- Security/access control

CLOUD COMPONENTS

- Clients / end-users
- Services
- Applications
- Platform
- Storage
- Infrastructure

Memory:

CSAPSI

Client - Service - Application

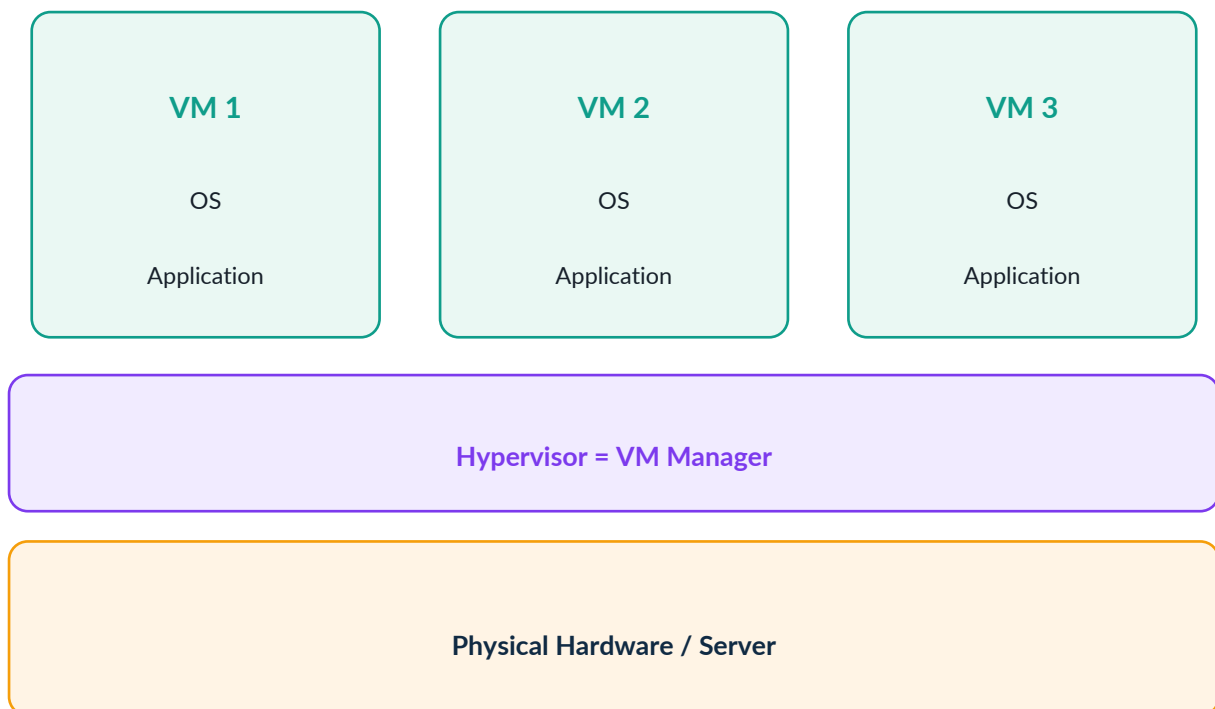
Platform - Storage - Infrastructure

7. Virtualization + Hypervisor

Easy diagram question

CORE IDEA

Virtualization allows one physical server to run multiple virtual machines (VMs). A hypervisor sits above the hardware and manages the VMs.



VIRTUALIZATION MEMORY

1 Physical Server -> Hypervisor -> Many VMs

8. Sensor-Cloud

Very important IoT-cloud integration topic

EASY DEFINITION

Sensor-Cloud virtualizes physical sensor resources so that the same sensor infrastructure can be offered to multiple users or applications in a cloud-style, pay-per-use model.

MAIN FEATURES

- Virtualization of sensor nodes/resources
- Pay-per-use idea
- One physical sensor/network can appear as many virtual resources
- Acts as a layer between physical sensors and end-users
- Sensor resources can serve multiple applications/users
- Reduces the need for every user to own a dedicated WSN

SENSOR-CLOUD MEMORY

VPMS = Virtualize - Pay-per-use - Many users - Stratum between sensors and users

9. WSN vs Sensor-Cloud + Actors

High-probability comparison question

Traditional WSN

- Often dedicated to one user
- WSN user handles more ownership/maintenance responsibilities
- Data typically serves a specific application
- Less virtualization/resource sharing

Sensor-Cloud

- Virtualized sensor resources
- Serves multiple users/applications
- Responsibilities distributed among actors
- Cloud-style resource sharing and pay-per-use

3 ACTORS

- End-user: uses Sensor-as-a-Service without needing to know the physical sensor.
- Sensor Owner: buys and deploys physical sensors, then lends them to Sensor-Cloud.
- SCSP: Sensor-Cloud Service Provider; manages services and charges users by usage.

ACTOR MEMORY

EOS = End-user - Owner - SCSP

10. Sensor-Cloud Architecture / Workflow

Memorize the flow, not every box

ROLE SUMMARY

- End-user registers, selects templates, and requests applications.
- Sensor owner deploys physical sensor nodes in different locations.
- SCSP plays the managerial/resource-allocation role.

1 User Request

2 Create Virtual Sensor

3 Allocate / Schedule Resource

4 Collect / Aggregate Data

5 Provision Data

WORKFLOW MEMORY

Request -> Virtualize -> Allocate -> Collect Data -> Provision -> Release

11. Mobile Cloud Computing (MCC) vs Edge

Know why mobile/IoT devices offload work

MCC - MOBILE CLOUD COMPUTING

- Mobile/IoT devices use powerful remote cloud computing and storage resources.
- Useful when local CPU is insufficient, data is large, or battery is limited.
- Computation offloading can extend battery life.
- Supports sophisticated applications and massive storage.

MCC MEMORY

BCS = Battery saving - Computing power - Storage

MCC

- Centralized
- Far from UE
- Longer latency
- Higher jitter
- Abundant compute/storage

EDGE

- Distributed
- Close to UE
- Short latency
- Lower jitter
- More limited compute/storage

12. Fog vs MEC vs Cloudlet

Use the location clue to identify the right answer

FOG

- Near-user edge devices
- Routers, switches, APs, gateways
- Between end devices and cloud
- Can support IP and non-IP access

MEC

- Cloud/IT services at mobile cellular edge
- Servers near base-station/mobile network
- Faster mobile cloud service
- Reduces network congestion

CLOUDLET

- Small-scale cloud data center
- Near mobile devices
- VM-based resource provisioning
- Often one-hop over Wi-Fi

IDENTIFY FAST

Router/Gateway -> Fog | Base station/Mobile network -> MEC | Data center in a box/Wi-Fi -> Cloudlet

13. Final 5-Minute Revision

Read this page three times before the exam

FACT BANK

- Cloud = shared + on-demand + rapid + anywhere
- NIST = BRMOR
- SaaS = use software
- PaaS = build/deploy apps
- IaaS = manage VM/OS/network/storage
- Deployment = PPHC
- Public = everyone; Private = one org; Hybrid = mix
- Virtualization = 1 physical server -> hypervisor -> many VMs
- Sensor-Cloud = virtualization + pay-per-use + many users
- Actors = EOS
- MCC = remote cloud for battery/computing/storage
- Edge = close to UE, short latency
- Fog = router/gateway near devices
- MEC = mobile cellular edge
- Cloudlet = data center in a box

ULTIMATE MEMORY CHAIN

SORA -> BRMOR -> SPI -> PPHC -> EOS -> BCS -> Fog/Edge/MEC/Cloudlet

14. Top Questions + 8-Mark Formula

What to prepare first tonight

TOP 10 QUESTIONS

- Define cloud computing and explain its characteristics.
- Explain five NIST essential characteristics.
- Compare SaaS, PaaS and IaaS with examples.
- Explain Public, Private, Hybrid and Community cloud.
- Explain virtualization and hypervisor.
- Explain Sensor-Cloud and its features.
- Compare WSN and Sensor-Cloud.
- Explain Sensor-Cloud actors and workflow.
- Explain Mobile Cloud Computing and compare it with Edge.
- Compare Fog, MEC and Cloudlet.

8-MARK ANSWER FORMULA

- 1. Write one-line definition.
- 2. Mention the exact model/technology.
- 3. Write 5-7 clean points.
- 4. Add one comparison or small flow if relevant.
- 5. Add example(s) where the slide gives them.
- 6. Finish with one-line conclusion.

LAST RULE

Memorize keywords + order. Do not try to memorize full paragraphs.